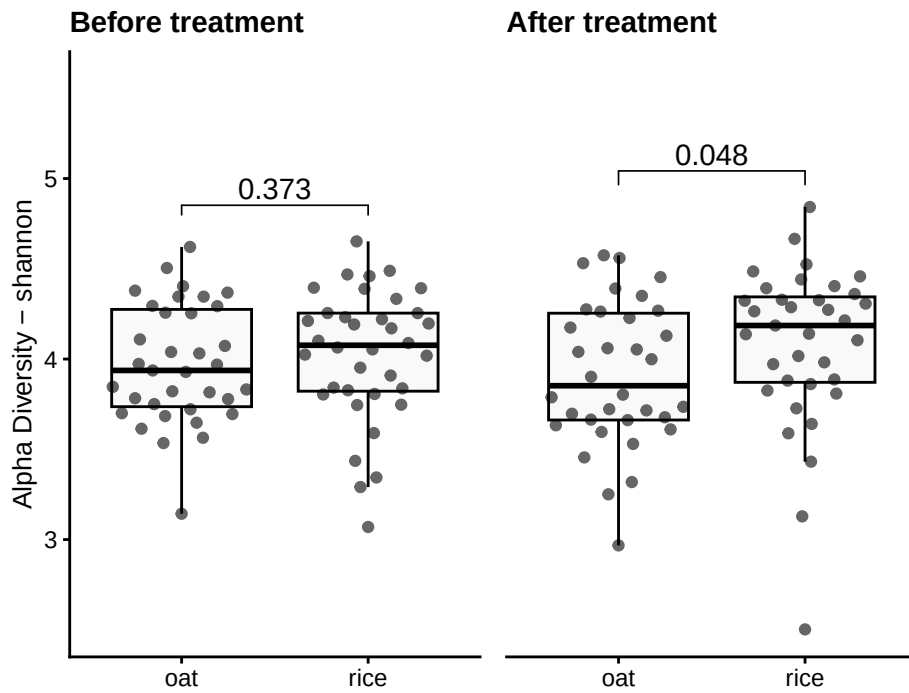


Alpha Diversity

Muluh

1 Group-wise comparisons

- **Index:** shannon
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



1.1 Table of the top findings

1.1.1 Across-group

1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	4	4	0.02	0.373

The alpha diversity between the ‘oat’ and ‘rice’ groups appears to be very similar, with minimal difference in the mean values (both 4.00).

The log2FC indicates almost no change between the two groups in terms of diversity (0.02), further supporting the notion of no substantial difference.

The p.adj of ‘0.373’ shows that the difference between the groups is not statistically significant. Thus, we would conclude that the treatment (‘oat’ and ‘rice’) does not have a significant impact on the alpha diversity of the groups at the baseline.

1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

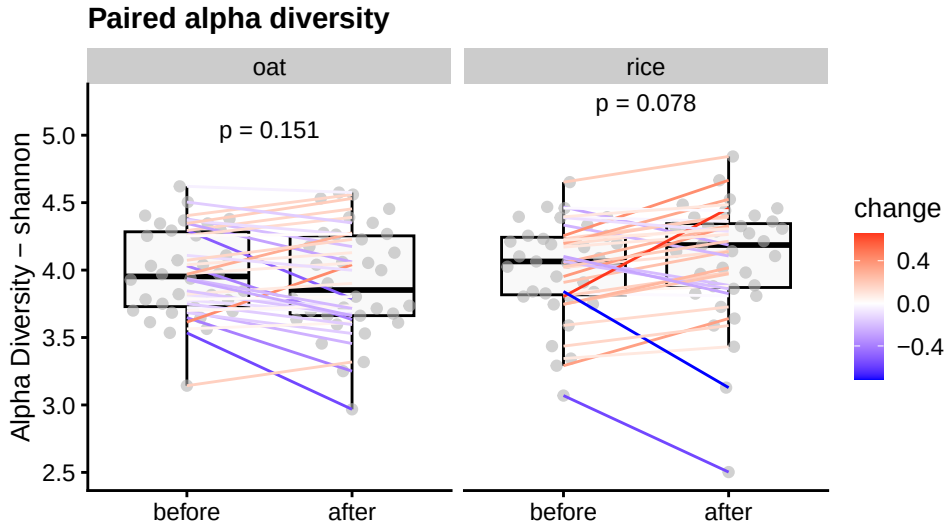
Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	3.9	4.1	0.06	0.048

After treatment, the mean alpha diversity for ‘oat’ is 3.90, and for ‘rice’ is 4.10, showing a slight difference between the two groups, with ‘rice’ having slightly higher diversity.

The log2FC value of 0.06 indicates a very small positive difference, suggesting that the ‘rice’ group experienced a slight increase in diversity compared to ‘oat’.

The p.adj of 0.0480 indicates that the difference between the groups is statistically significant (since it’s less than 0.05), meaning that the observed difference in diversity between the ‘oat’ and ‘rice’ groups after treatment is likely not random.



1.1.2 Within-group

1.1.2.1 Paired treatment

- Multiple testing correction: **paired -none**
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
oat	4	3.9	-0.02	0.151
rice	4	4.1	0.03	0.078

For the oat diet, there was a slight decrease in alpha diversity from Timepoint 1 (mean = 4.00) to Timepoint 2 (mean = 3.90), but the change was not statistically significant (p-value = 0.151).

For the rice diet, there was a slight increase in alpha diversity from Timepoint 1 (mean = 4.00) to Timepoint 2 (mean = 4.10), but again, the change was not statistically significant (p-value = 0.0780).

1.2 LMM

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**
- Formula used: $\sim \text{diet} * \text{timepoint} + (1 | \text{id})$

Table 4: Table of interaction effects

Term	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	4.035	0.089	135.874	45.491	0.000
dietrice	-0.076	0.125	135.874	-0.607	0.545
time	-0.062	0.043	67.447	-1.464	0.148
dietrice:time	0.127	0.060	67.442	2.119	0.038

(Intercept):

Estimate: **4.04** This represents the baseline value of the alpha diversity when both the diet and timepoint are at their reference categories (oat for diet and before for timepoint). **Pr(>|t|): 4.19e-84**, which is extremely small and indicates highly significant evidence that the intercept term is different from zero.

dietrice:

Estimate: **-0.08** This is the difference in alpha diversity between rice and the reference diet (oat), after adjusting for the effect of time. **Pr(>|t|): 0.545** suggests that the diet (rice vs oat) effect is not statistically significant. The difference in alpha diversity between rice and oat diets is not substantial enough to conclude that it is different.

time:

Estimate: **-0.06** This is the difference in alpha diversity between timepoints (after vs before), after adjusting for the effect of diet. **Pr(>|t|): 0.148** indicates that the effect of time (before vs after) on alpha diversity is not statistically significant. This suggests that alpha diversity does not change significantly between timepoints in the study.

dietrice:time (Interaction):

Estimate:**0.13** This is the interaction between diet (rice vs oat) and timepoint (before vs after). The positive estimate suggests that the effect of time on alpha diversity differs between the rice and oat diets. **Pr(>|t|): 0.0378** indicates that the interaction between diet and timepoint is statistically significant at the usual significance level (0.05). This suggests that the relationship between alpha diversity and timepoint differs for the rice and oat diets.

2 Alternate analyses

- **Beta (Estimate):** Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

3 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(15): sample id ... shannon group
```

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.065
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.20, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	3.5	3.1, 3.9	<0.001
meal_group			0.3
1	—	—	
2	0.23	-0.07, 0.53	0.14
3	0.16	-0.14, 0.46	0.3
4	0.13	-0.05, 0.31	0.2
age	0.01	0.00, 0.01	0.039

¹CI = Confidence Interval

```

reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```